

AMR KHALAFALLA

AI & Data Scientist

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PROFESSIONAL SUMMARY

Results-driven AI & Data Scientist with expertise in Machine Learning, Computer Vision, and Natural Language Processing. Proven track record of designing and deploying scalable AI solutions using Python, PyTorch, TensorFlow, and AWS ML services. Specialized in Retrieval-Augmented Generation (RAG) systems and real-time computer vision applications. Demonstrated ability to translate complex business requirements into innovative AI-powered solutions that drive measurable outcomes.

TECHNICAL SKILLS

Programming & Frameworks: Python, Flask, Django, LangChain, SQL, REST APIs

AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), Neural Networks, Model Optimization

ML Libraries & Tools: PyTorch, TensorFlow, Scikit-learn, OpenCV, FAISS, YOLO, FaceNet, Keras, Pandas, NumPy

Cloud & Deployment: Amazon SageMaker, AWS Machine Learning Services, Model Deployment, API Development

Data Engineering: Data Preprocessing, Feature Engineering, Web Scraping (Selenium), Database Management (SQLite)

Professional Skills: Team Leadership, Cross-functional Collaboration, Technical Communication, Problem Solving

PROFESSIONAL EXPERIENCE

AI & Data Science Trainee | Digilians Initiative

August 2024 – July 2025

- Completed intensive 9-month applied AI and Data Science program, mastering end-to-end machine learning workflows from data collection to model deployment
- Engineered and optimized 15+ machine learning models achieving 85%+ accuracy for classification and prediction tasks using scikit-learn, TensorFlow, and PyTorch
- Architected production-grade NLP pipelines and Retrieval-Augmented Generation (RAG) systems using LangChain and FAISS, improving information retrieval accuracy by 40%
- Collaborated in agile, cross-functional teams of 6+ members, delivering projects 20% ahead of schedule through effective sprint planning and daily standups

KEY PROJECTS

ARGUS AI – AI-Powered Exam Proctoring System | Graduation Project

June 2025

- Designed and implemented real-time AI-based exam monitoring system detecting 12 types of cheating behaviors with 92% accuracy, processing 30 FPS video streams
- Integrated YOLOv8 object detection model for unauthorized device identification achieving 89% precision and FaceNet for facial recognition with 95% verification accuracy
- Developed custom mobile phone detection pipeline using OpenCV and TensorFlow, reducing false positives by 35% through advanced image preprocessing techniques
- Built full-stack monitoring dashboard using Django with SQLite database, featuring real-time alerts, comprehensive reporting, and RAG-powered natural language query interface

Technologies: Python, OpenCV, TensorFlow, YOLOv8, FaceNet, Django, SQLite, LangChain, REST APIs

AI-Powered Job Search Platform

September 2025

- Engineered automated web scraping pipeline using Selenium, extracting and updating 500+ job listings daily from 20+ recruitment websites
- Implemented intelligent CV parsing system using NLP techniques, automatically extracting 25+ skill categories with 91% accuracy
- Architected RAG-based career advisor using LangChain and FAISS vector database, providing personalized career guidance with 85% relevance score
- Deployed semantic similarity search algorithm achieving 0.82 cosine similarity score for job-candidate matching, increasing successful placements by 45%

Technologies: Python, Flask, LangChain, FAISS, Selenium, NLP

Character Simulator – RAG-Based Conversational AI

April 2025

- Built multi-character conversational AI system supporting 8 distinct personalities with context-aware dialogue generation using RAG architecture
- Implemented FAISS vector database for semantic search across 50,000+ dialogue samples, achieving 200ms average query response time
- Designed conversation memory tracking system maintaining context across 10+ dialogue turns with 88% coherence score

Technologies: Python, LangChain, FAISS, Vector Databases, Prompt Engineering, RAG

EDUCATION

Bachelor of Artificial Intelligence

Faculty of Artificial Intelligence, Menoufia University

Expected Graduation: 2025 | Overall Grade: Good

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Aspire Leadership Program** – Harvard Business School Online | June 2024 – August 2024
- Completed 30-hour leadership development program with 150+ international participants, focusing on strategic decision-making and team management
- **Odoo ERP Training Program** | June 2024 – August 2024
- Gained hands-on experience in enterprise resource planning covering finance, sales, inventory management, and AI integration within ERP systems
- **Artificial Intelligence Technical Training** – National Telecommunication Institute (NTI) | June 2022 – August 2022
- Completed 90 hours of technical AI training and 30 hours of professional skills development covering ML algorithms, neural networks, and industry best practices

LEADERSHIP & ACTIVITIES

Public Relations Coordinator

- Coordinated event logistics and stakeholder communications, achieving 95% participant satisfaction rate through effective planning and execution

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency